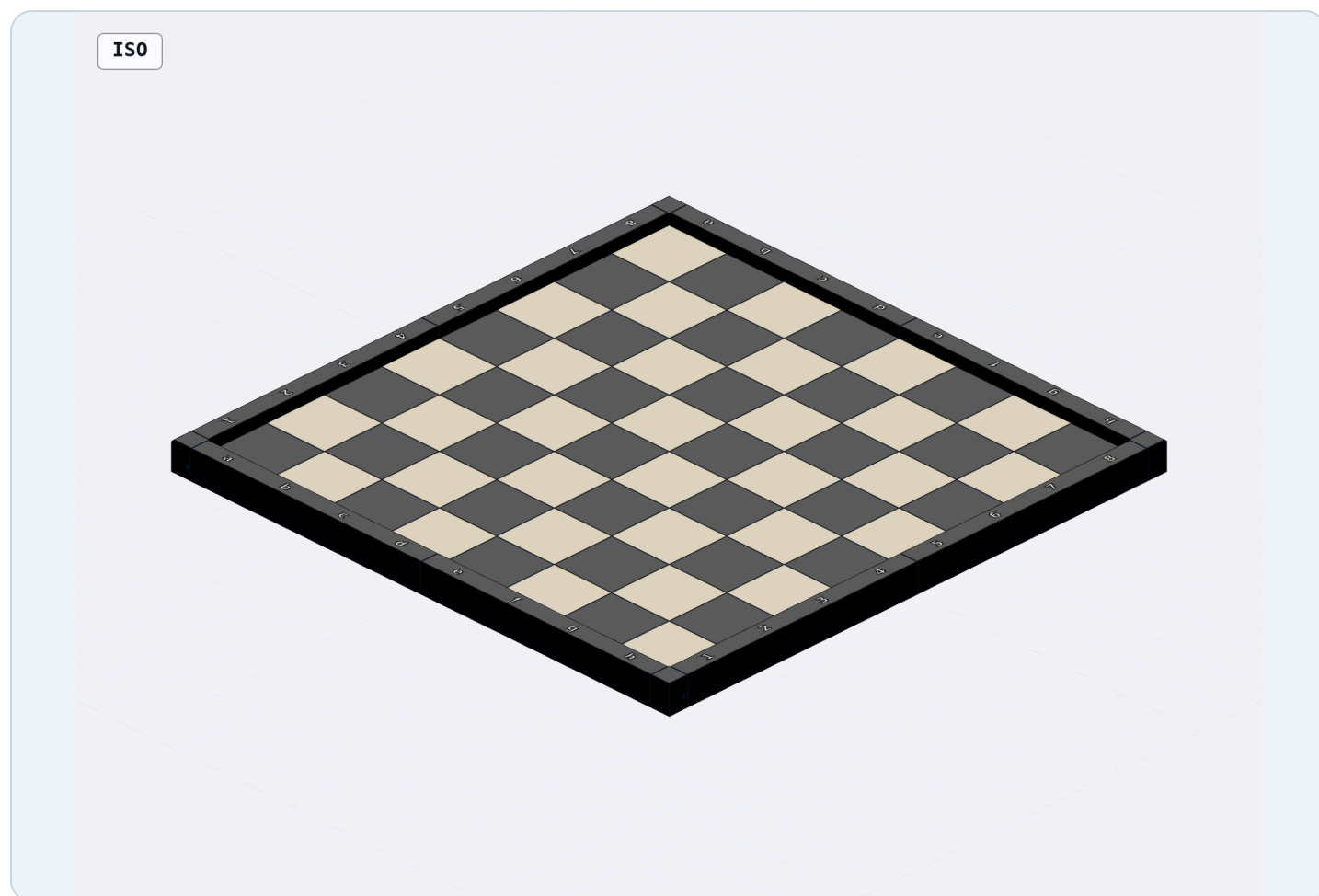


60 mm Modular Chessboard

Four-quarter, full-size board for the Bambu Lab X2D, with raised perimeter, contrasting algebraic notation, concealed seam registration, and screw-retained corner caps.



60 mm

square size

480 mm

playing area

520 mm

overall size

10 mm

perimeter rise

Read before assembly. Several nuts become inaccessible after adjoining parts are installed. Follow the order in this manual and complete every “load nut first” step before closing a seam or adding a corner cap.

Designed for PLA Matte. Hardware: M3×12 and M2×12 screws with standard hex nuts. No heat-set inserts are required.

Parts, hardware, and preparation

Printed parts

PART	QTY.
Single-color playing-quarter bodies	4
Named perimeter rails	8
Underside seam bridges	8
Corner caps	4
Glue-in dark-square inlays	32
Notation insert set, separate-print route	1

Hardware

HARDWARE	QTY.
M3×12 screws	32
Standard M3 hex nuts	32
M2×12 screws	4
Standard M2 hex nuts	4
Adhesive felt/rubber pads	12

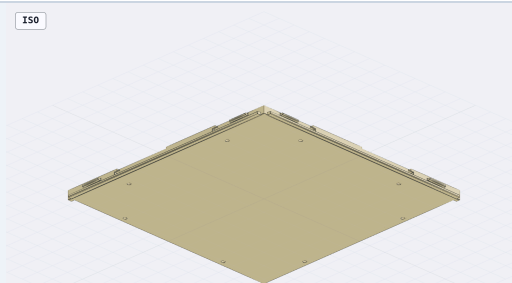
Nominal nuts: M3 = 5.5 mm across flats × 2.4 mm thick; M2 = 4.0 mm across flats × 1.6 mm thick.

Useful tools

- Drivers matching your M3 and M2 screw heads
- Tweezers or needle-nose pliers for nut placement
- Deburring tool, hobby knife, or small flat file
- Protected, flat work surface and soft cloth
- PLA-compatible adhesive for separate inlays

Identify the faces

- **Playing side:** checker pattern and raised border
- **Underside:** screw recesses, nut entrances, bridges
- **Internal edges:** long lower tongue/groove plus short upper keys
- **Outer edges:** grooves that receive rail tongues



Color parts

Print each **playing_surface_body** in the chosen light color. Print the 32 **dark_square_inlay** parts on the two dedicated dark-color plates. Dry-fit every square, then apply a thin, even layer of PLA-compatible adhesive and press it fully into its recess until flush. Keep each rail as one object and assign its inset notation a contrasting filament.

Clean fit surfaces

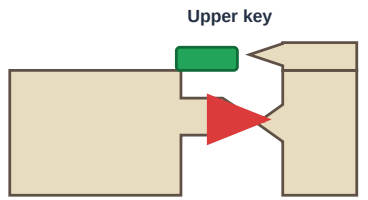
Remove brim material, strings, and burrs. Preserve the 0.5 mm lead-ins and 0.4 mm first-layer relief. Never force a connector that has visible debris.

Hardware trial: Test one M3 nut and one M2 nut before loading all pockets. A nut should slide into its channel, settle into the hex pocket, and remain unable to rotate.

Join the four playing quarters

Quarter joint — align, slide, and verify the playing faces

1 Align the profiles

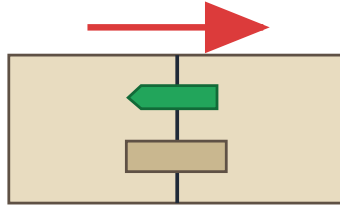


Long lower tongue
carries shear load

Matching grooves

Keep both quarters fully supported and level.

2 Slide straight together

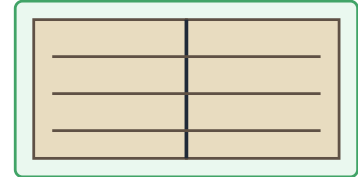


Do not twist or pull with screws

The 0.5 mm lead-ins start the fit; the 0.25 mm nominal clearance permits sliding.
Stop and clean debris if the joint binds.

3 Verify the locked alignment

Continuous, flush playing plane



Upper keys prevent a visible lip

The underside bridges installed next provide the final clamping and bending stiffness.
No raised square edges should be visible.

The long lower tongue carries shear; the two short upper keys register the playing faces flush.

- 1 **Work face-down.** Place all four quarters on a clean, padded, flat surface in their SW, SE, NW, and NE positions.
- 2 **Load all M3 nuts before joining.** Each quarter has eight side-loading nut entrances—two per edge. Slide one standard M3 nut into every entrance until it seats in the internal hex pocket over the vertical screw passage.
- 3 **Join the south row.** Slide the southwest quarter's east tongues and upper keys into the southeast quarter's west grooves. Keep the faces supported and move the parts straight together.
- 4 **Join the north row.** Slide northwest into northeast using the same method.
- 5 **Join the two rows.** Slide the completed north row southward onto the north-facing tongues of the south row. The two short upper keys on each edge register the playing faces while the long lower tongue carries shear load.
- 6 **Check the top seam.** Sight across the playing face while supporting the full assembly. The quarter faces should be level with no trapped filament or visible step.

Designed clearance: the lower tongue/groove and concealed upper keys use 0.25 mm nominal depth clearance. A smooth sliding fit is expected; a press fit is not.

Do not force the joint. If the quarters stop before the seam closes, separate them and remove brim, stringing, or debris from both profiles.

Lock the quarter seams from below

Underside seam bridge — both screws pull against captured M3 nuts

1 Place the bridge across the seam

Preloaded M3 nuts in the quarter edges



Recessed screw-head face points outward/down

Center the bridge over the seam and align both holes.

Start both M3×12 screws before tightening either.

2 Alternate until seated — then stop

Quarter seam



Locked state

Bridge sits flat; screw heads remain recessed; seam is closed.

Tighten left → right → left only until the board is flat.

Vertical center seam

Install four bridges centered at Y = -180, -60, 60, and 180 mm.

Horizontal center seam

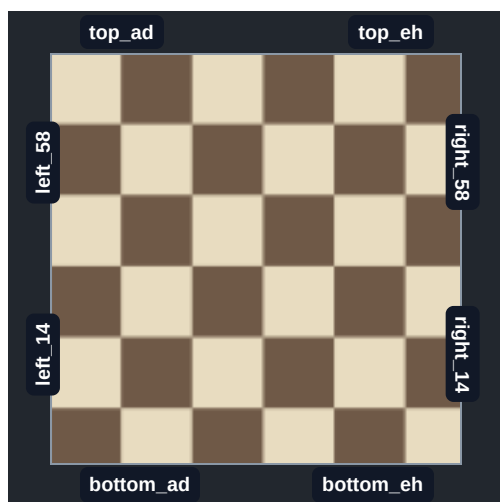
Install four bridges centered at X = -180, -60, 60, and 180 mm.

- 1 **Keep the board face-down and fully supported.** Confirm the correct M3 nuts were loaded before the seams were closed.
- 2 Place each bridge across its seam with the two recessed screw-head seats facing outward/downward.
- 3 Insert two M3×12 screws through the bridge and into the captured nuts. Turn each screw only a few threads at first.
- 4 Alternate between the two screws until the bridge sits flat. Repeat for all eight bridge locations.
- 5 Sight across the playing side after supporting the full board. Loosen and reseal any bridge associated with a visible seam lip.

How the bridge locks: each screw pulls against a nut captured inside the quarter edge. The bridge spans both panels, clamps the seam closed, and resists bending without any exposed hardware on the playing side.

Stop when seated. Excess preload can bow the 240 mm panels, crush printed layers, or cause long-term PLA creep.

Install the raised perimeter rails



White sits at the bottom a–h side. The a1 square must be dark.

Rail positions

White side	bottom_ad, bottom_eh
Opposite side	top_ad, top_eh
Left from White	left_14, left_58
Right from White	right_14, right_58

Notation check

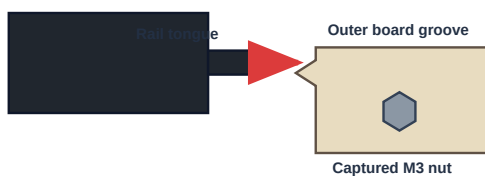
Letters and numbers should be upright when viewed from their nearby player position. Adjacent rail segments meet under a separate corner cap.

- 1 With the board still face-down, place each rail beside its matching quarter edge.
- 2 Align the rail's long tongue with the quarter's outer groove. Start at one end and slide the rail straight into place; keep the screw passages aligned.
- 3 Install two M3×12 screws from the underside of each rail into the previously loaded M3 nuts. Start both screws before tightening.
- 4 Repeat for all eight rails. Leave the four corner caps off until the M2 lock nuts are loaded in the next stage.

Perimeter rail — slide the tongue into the board, then fasten from below

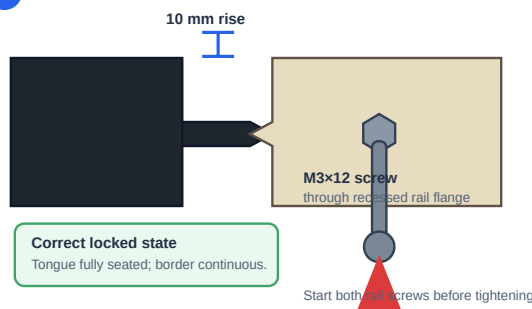
1 Engage the full-length rail tongue

Black raised rail



Slide the rail straight along the entire edge.
Do not use the screws to pull a blocked tongue into place.
The 0.5 mm lead-in should start the connection easily.

2 Fasten the seated rail from below



Start both screws before tightening.

Correct fit

The rail tongue seats fully, the 20 mm border is continuous, and the perimeter top remains 10 mm above the playing surface.

If a rail stops early

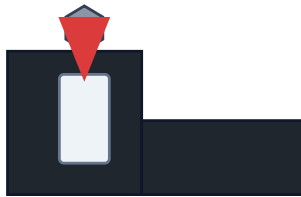
Remove it, inspect the entrance and tongue, then trim only obvious brim or stringing. Do not draw a misaligned rail home with the screws.

Maximum printed length: each rail is 247.05 mm long. If reprinting, position a 3–4 mm brim carefully inside the X2D's 256 mm main-nozzle area.

Lock the four corner caps

Corner-cap lock — captured nut, downward cap motion, cross-screw retention

1 Drop in the M2 nut



Use the named locking rib

SW bottom_ad • SE right_14
NE top_eh • NW left_58

Nut must lie flat and cannot rotate.

2 Slide the cap downward



Cap slots capture both adjacent ribs

Keep the cap level so both ribs enter together.
Rotate the recessed side hole toward the selected rib before sliding the cap down.

3 Install the cross-screw

Screw crosses cap wall and rail rib



Locked state

M2x12 prevents the cap lifting off.

Tighten only until the cap is retained.

Load one M2 nut at each corner

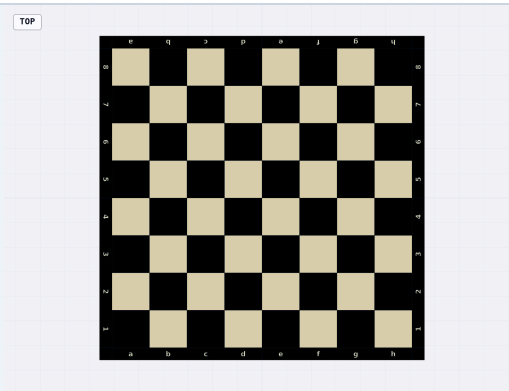
CORNER	RAIL RIB THAT RECEIVES THE NUT	CAP SCREW DIRECTION
Southwest	bottom_ad	From the nearby outside face
Southeast	right_14	From the nearby outside face
Northeast	top_eh	From the nearby outside face
Northwest	left_58	From the nearby outside face

- 1 Confirm the M2 nut lies flat in the hex pocket and its threaded hole aligns with the horizontal passage through the rail rib.
- 2 Rotate the identical corner cap so its recessed cross-hole faces the selected rib and is accessible from the nearby outside edge.
- 3 Slide the cap down over both ribs. The shallow slot lead-ins help both ribs enter together; keep the cap level.
- 4 Insert one M2×12 screw through the cap wall and rail rib, then thread it into the captured nut.
- 5 Tighten only until the screw head sits in the 4.5 mm diameter × 2.2 mm deep recess and the cap cannot lift.
- 6 Repeat at all four corners.

How the lock works: the M2 nut is captured against rotation in the rail rib. After the cap is installed, the cap roof also prevents the nut from escaping upward. The cross-screw positively prevents the cap from lifting off.

Do not substitute a longer screw without checking. M2×12 is the validated length. A longer screw may bottom out or damage the printed pocket.

Final setup, inspection, and care



Final checklist

- | | |
|--|--|
| ☐ All 8 seam bridges installed with 16 M3 screws | ☐ All 8 rails installed with 16 M3 screws |
| ☐ All 4 caps locked with M2×12 screws | ☐ No screw head protrudes below a recessed seat |
| ☐ Playing-surface seams are flush | ☐ a1 is dark and notation is correctly oriented |
| ☐ Rails are fully seated and border joints are even | ☐ No loose nuts rattle inside the assembly |
- 1 Apply eight felt/rubber pads to the rail recesses and four to the corner-cap recesses.
 - 2 With assistance if needed, support the full 520 mm board and turn it playing-side up. Do not lift it by a single rail or corner cap.
 - 3 Place it on a flat surface and check for rocking. If needed, verify pad thickness and loosen/reseat a bridge before retightening evenly.
 - 4 After the first several uses, check that the screws remain snug. PLA does not benefit from high fastener preload.

Tight connector

Remove the part, clean the entrance, and lightly deburr only the interfering edge. Do not hammer the quarters, rails, or caps together. If consistently tight after reprinting, use small slicer XY-hole/contour compensation.

Visible seam lip

Check for debris in the upper-key grooves, verify the assembly is on a flat surface, and tighten opposing bridge screws gradually. The keys register height; the bridges establish final stiffness.

Cap screw will not start

Remove the cap and confirm it was rotated toward the selected locking rib. Reseat the M2 nut flat in its top-loaded hex pocket and try again without forcing the thread.

Routine care

Clean with a dry or lightly damp cloth. Avoid hot cars, heaters, direct prolonged sun, solvents, and soaking. Store the large panels supported flat to minimize PLA creep.

Design reference: STEP/STL files and the latest validation report are in the project folder. The generated assembly measures 520 × 520 × 27.6 mm and contains 60 mm squares over a 480 × 480 mm playing area.

Assembly complete. Tighten only enough to keep seams closed and the board flat.